



NHS OUTPATIENT PERFORMANCE ANALYSIS REPORT

Content:

A Data-Driven Optimization Study Final Project Report Analyzed 113 Million Appointments Identified £18-30M in Opportunities 87K-114K Additional Capacity Available

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EXECUTIVE SUMMARY

OVERVIEW

This project analyzed 113 million NHS outpatient appointments to identify strategic opportunities for capacity expansion and efficiency improvement.

CURRENT SITUATION

The NHS outpatient system faces three critical challenges:

1. **Data Quality Gap** - 92% of appointments lack clear diagnosis codes (104M of 113M)
2. **Follow-Up Inefficiency** - Follow-ups vary wildly (0.7 to 14.1 per patient)
3. **Telehealth Underutilization** - Only 17.99% remote despite 40-45% clinical suitability

THREE OPPORTUNITIES IDENTIFIED

Opportunity 1: Diagnosis Clarity (79% → 95%)

- Appointments Freed: 15K-30K annually
- Savings: £5-8M per year
- Investment: £20-50K
- Timeline: 6 months

Opportunity 2: Follow-Up Optimization

- Appointments Freed: 22K-44K annually
- Savings: £8-12M per year
- Investment: £50-100K
- Timeline: 6-12 months

Opportunity 3: Telehealth Expansion (17.99% → 45%)

- Additional Capacity: 50K+ appointments annually
- Savings: £7-14M per year
- Investment: £100-200K
- Timeline: 12 months

TOTAL OPPORTUNITY

- Additional Appointments: 87K-114K (6.6-8.7% capacity increase)

- Annual Savings: £18-30M
- Total Investment: £295K
- Return on Investment: 61-102x (Year 1)
- Payback Period: Month 1-2

STRATEGIC APPROACH:

Three-phase Agile implementation over 12 months:

Phase 1 (Q1-Q2): Quick Wins

- Fix top 2-3 data quality issues
- Pilot telehealth with 5 proven-safe diagnoses
- Standardize top 5 diagnosis pathways
- Expected impact: +20K appointments

Phase 2 (Q3-Q4): Scale

- Expand to 25 telehealth diagnoses
- Implement hybrid pathways for 20+ diagnoses
- Automated follow-up monitoring
- Expected impact: +35K appointments

Phase 3 (Year 2): Optimize

- Remote monitoring technology integration
- AI-assisted triage
- Full digital-first approach
- Expected impact: +50K appointments

WHAT WE NEED

✓ Executive Sponsor (monthly steering committee) ✓ Clinical Lead (25% time) ✓
Budget: £295K over 12 months ✓ Authority to pilot changes ✓ IT support for platform readiness

EXPECTED OUTCOMES (12 Months)

- 87K-114K additional appointments
- £18-30M in cost savings
- Patient satisfaction increased to 85%+

- System capacity from 113M to 127M+ appointments
- Data quality from 79% to 95% diagnosis coding
- Telehealth adoption from 17.99% to 40-45%

RECOMMENDATION

Proceed with Phase 1 quick wins immediately. Validate approach through pilots, then scale to Phases 2-3. Expected positive ROI from month 1.

PROJECT OVERVIEW

PROJECT NAME NHS Outpatient Performance Analysis & Optimization Initiative

OBJECTIVES Analyze 113M appointments, identify inefficiencies, quantify improvements, and develop implementation roadmap.

SCOPE - IN SCOPE

- Analysis of 113M appointments
- 89M diagnoses (plus 104M unknowns)
- Attendance patterns (first, follow-up, telehealth)
- Diagnosis-specific metrics
- Telehealth suitability scoring
- Interactive Power BI dashboard
- Agile implementation roadmap
- Financial modelling and ROI

SCOPE - OUT OF SCOPE

X Inpatient data

X Emergency department

X Implementation execution (planning only)

X Clinical protocol changes

METHODOLOGY

Step 1: Data Processing (Python)

- Extract 113M records
- Validate and clean data
- Create analytical datasets

Step 2: Exploratory Analysis (Python)

- Distribution by diagnosis, type, modality
- Trend analysis (6+ months history)
- Correlation analysis

Step 3: Diagnostic Analysis

- Follow-up ratios per diagnosis
- Priority scoring
- Telehealth suitability scoring

Step 4: Visualization (Power BI)

- Executive summary dashboard
- Diagnosis demand analysis
- Operational efficiency metrics
- Telehealth opportunity mapping

Step 5: Strategic Recommendations

- Agile implementation framework
- Risk mitigation
- Success metrics
- Phase-gate approach

DATA SOURCES

- NHS Outpatient Records Database
- Period: [Your dates]
- Coverage: [Locations/Trusts]
- Records: 113 million

KEY ASSUMPTIONS

- Historical trends continue
- Telehealth clinical suitability validated
- Staff time savings based on NHS benchmarks
- Patient adoption of telehealth >80%

- No major external disruption

KEY FINDINGS

FINDING 1: DATA QUALITY CHALLENGE "92% of Appointments Lack Clear Diagnosis Codes"

Current State: 79% coded (89M of 113M) Problem: 104M appointments lack diagnosis clarity Root Causes: Intake gaps, coding inconsistency, system limitations Impact: 15K-30K unnecessary appointments annually

Opportunity: Improve to 95% diagnosis clarity Actions:

1. Root cause analysis (why are diagnoses missing?)
2. Implement pre-visit questionnaire
3. Clinician coding training
4. System prompt for diagnosis

Investment: £20-50K Timeline: 6 months Savings: £5-8M annually Benefit: Enables all other improvements

FINDING 2: FOLLOW-UP PATHWAY INEFFICIENCY "Follow-Ups Vary from 0.7 to 14.1 Repeats Per Patient"

Current State: Average 1.6 follow-ups (range: 0.7-14.1) Problem: Extreme variation suggests some diagnoses have too many/few follow-ups Root Causes: No standardized pathways, unclear discharge criteria, defensive medicine Impact: Estimated 15-20% of follow-ups are unnecessary

Opportunity: Evidence-based follow-up protocols Actions:

1. Clinical consensus on ideal follow-ups per diagnosis
2. Automate routine monitoring (replace with remote checks)
3. Clear discharge criteria
4. Alerts for deviations from protocol

Investment: £50-100K Timeline: 6-12 months Savings: £8-12M annually Benefit: 22K-44K appointment slots freed

FINDING 3: TELEHEALTH UNDERUTILIZATION "Only 17.99% Remote Despite 40-45% Potential"

Current State: 17.99% of 113M = 20M remote appointments Problem: 50+ diagnoses at 0% despite clinical suitability Barriers: Uncertainty, technology limits, patient preference, clinician comfort

Current Success Examples:

- ✓ Actinomy cetoma: 100% remote
- ✓ Acute bronchitis: 100% remote
- ✓ These prove feasibility for low-acuity conditions

Opportunity: Phase expansion to 45% adoption Phase 1 (Q1-Q2): Scale 10 → 25 diagnoses at 100% remote Phase 2 (Q3-Q4): Hybrid models for 15 diagnoses (50% remote) Phase 3 (Year 2): Full integration with remote monitoring

Investment: £100-200K Timeline: 12 months Savings: £7-14M annually Benefit: 50K+ additional appointments

FINANCIAL SUMMARY

	Opportunity Freed	Savings	Investment	ROI
Data Quality	15-30K	£5-8M	£20-50K	100-400x
Follow-Ups	22-44K	£8-12M	£50-100K	80-240x
Telehealth	50K+	£7-14M	£100-200K	35-200x
TOTAL	87-114K	£18-30M	£295K	61-102x

RECOMMENDATIONS

RECOMMENDATION 1: DATA QUALITY IMPROVEMENT

What: Achieve 95% diagnosis clarity (from 79%)

Why: Foundation for all other improvements; enables optimal routing

How (Quick Wins): Week 1-2: Root cause analysis Week 3-6: Implement top 2 process fixes Week 7-12: Monitor and refine

Actions:

- ✓ Why do 104M lack diagnoses? (audit)
- ✓ Pre-visit questionnaire implementation
- ✓ Clinician training on coding

- ✓ Weekly measurement and feedback

Investment: £20-50K Timeline: 6 months Expected Savings: £5-8M/year

RECOMMENDATION 2: FOLLOW-UP PATHWAY STANDARDIZATION

What: Evidence-based protocols for top 20 diagnoses

Why: Extreme variance (0.7-14.1) suggests inefficiency

How: Month 1: Clinical consensus on ideal follow-ups Month 2-3: Automate routine monitoring Month 4: Pilot with 2 diagnoses Month 5-6: Roll out to remaining 18

Actions:

- ✓ Clinician workshops (evidence-based guidelines)
- ✓ Standardized protocols per diagnosis
- ✓ Remote monitoring for routine checks
- ✓ Clear discharge criteria
- ✓ Monthly variance monitoring

Investment: £50-100K Timeline: 6-12 months Expected Savings: £8-12M/year

RECOMMENDATION 3: TELEHEALTH EXPANSION ROADMAP

What: 3-phase rollout (17.99% → 45% adoption)

Why: 40-45% of diagnoses clinically suitable; currently underutilized

How:

PHASE 1 (Q1-Q2): Scale from 5 → 15 diagnoses

- Expand proven-safe diagnoses
- 5K-10K additional remote appointments
- Investment: £20K

PHASE 2 (Q3-Q4): Launch hybrid models

- 50% remote + 50% in-person for 15 diagnoses
- 30K additional remote appointments
- Investment: £100K

PHASE 3 (Year 2): Full optimization

- Remote monitoring technology

- AI-assisted triage
- 50K+ additional remote appointments
- Investment: £75K

Actions: ✓ Clinical safety validation ✓ Workflow redesign ✓ Platform upgrades ✓ Staff training ✓ Patient communication

Investment: £195K total Timeline: 12 months Expected Savings: £7-14M/year